Towards an Agency-centered Ontology of Game Mechanics

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ABSTRACT

Game mechanics, the affordances they create for players, and the agency experienced by players as a result of interacting with those mechanics are all, indisputably, intertwined. This work seeks to explore the relationship between game mechanics and agency by way of building an agency-centered ontology of video game mechanics. Building on existing framings of agency, we devise a set of agency-centered domains, properties belonging to those domains, and questions about game mechanics that reveal those properties. We build a dataset of game mechanics from different games and, using our domain-specific set of questions, illuminate properties of those game mechanics. Finally, we explore how those properties are interrelated, and identify examples of game mechanics that, according to our analysis, inspire the most agency.

CCS CONCEPTS

• **Applied computing → Media arts**.

KEYWORDS

video games, game mechanics, agency, ontology

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1 INTRODUCTION

If one were asked to describe a video game, they might answer with a list of things that the player can do within the game world, a description of the game world or the story unfolding in the world, how their actions affect these things, and whether or not they felt it had meaning to them. In other words, they would give an account of the game's mechanics, and the agency they felt while playing. Members of our community may recognize the words "mechanics" and "agency" as well-debated terms–a point that receives treatment in our discussion of adjacent work.

Definitions aside, why is agency something to be desired? Developers often have the explicit goal of allowing for as much agency as possible in their games. A player experiencing agency results in an objectively positive outcome: expectations that the player had were met, and any time and effort in playing the game mattered to them. But what are the criteria by which mechanics are designed to have agency? Are there individual agency-inspiring properties of mechanics that can be identified such that we can build a reasonable ontology? In this work, we draw upon the many definitions of both terms put forth by our colleagues to inform the creation of such an ontology.

In particular, we answer the following research questions: 1) How do we define agency-centered domains of mechanics, and what questions can we ask that best reveal properties of mechanics related to player agency? 2) How are these domains and properties interrelated, and are there domains or properties that contribute more to player agency than others? 3) What are extant examples of game mechanics that, according to our analysis, inspire the most player agency? Building upon the work of our colleagues, we devise a set of agency-centered domains, properties belonging to those domains, and questions about game mechanics that reveal those properties. We build a dataset of game mechanics from different games and, using our domain-specific set of questions, illuminate properties of those game mechanics. Finally, we explore how those properties are interrelated, and identify examples of game mechanics that, according to our analysis, inspire the most agency.

2 RELATED WORKS

Researchers in our communities have defined and studied game mechanics in myriad ways. The definitions used for the term "mechanic," as pointed out by Lo et al. [6], are so numerous as to introduce critical communication problems across our interconnected areas of research. Some view game mechanics through a programmatic lens, referring to them as "methods invoked by agents" [11]; "particular components of the game, at the level of data representations and algorithms" [3]; or "rule-based systems/simulations that encourage a user to explore and learn the properties of their possibility space." [2] Others apply a more design-centered definition, referring to game mechanics as "the guts of a design document" [9], "means to guide the player into particular behavior" [4], or essential gameplay activities that players perform repeatedly [10].

Similarly, the concept of agency has been variably characterized by members of our communities. In their examination of agency with regards to narrative-focused gameplay, Carstensdottir et al. [1] broadly define agency as "the phenomenon where a player feels that the actions presented to them in the context of the game are meaningful and that their choice of action has a meaningful impact on the context in which they are engaging." It has also been called "the satisfying power to take meaningful action and see the results of our decisions and choices" [7]; an effect that occurs "when the

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actions players desire are among those they can take (and vice versa) as supported by an underlying computational model" [8]; and "communication through meaningful mappings between player and the game system" [12].

All of these characterizations, for both terms, are valid and valuable. They describe the many facets that are necessary to truly understand these concepts. This work seeks not to determine which of these terms are appropriate or lacking; but rather, to use the inherent relationships between them in synthesis of a model that can be validated. To our knowledge, an ontology of this nature, linking agency directly to properties of game mechanics, has not been put forth by our communities.

3 METHODOLOGY

3.1 Defining Agency-centered Domains

Characterizations of agency put forth by our colleagues often refer to a player having actions available to them that they desire to take, the action itself performing a desired function, and the meaning associated with the outcome of that action. We map these qualities of agency to three domains: availability, utility, and impact, respectively. We then explored these domains with probing questions focused on a given game mechanic. We describe these domains, the questions asked, and the purpose of these questions in the following sections. In the interest of brevity and legibility of our later statistical analysis, we use the identifiers A1, U1, I1, etc. to describe the domain from which the property comes, and the numerical assignment of the property, in order of their appearance.

3.1.1 *Availability.*

- *A1: Is this mechanic imposed upon the player?* This question seeks to differentiate between mechanics that the player must always choose to interact with, and the mechanics that the player is forced to interact with.
- *A2: Can this mechanic be used at will?* This question seeks to differentiate between mechanics that the player can always choose to interact with when they desire it, and mechanics that are only available in certain contexts.
- *A3: Is this mechanic unrestricted by other mechanics?* Even if the mechanic can be interacted with at will, its use may be restricted by other mechanics (e.g., choosing between two actions given that the player can only choose one).
- *A4: Can this mechanic be used from the very beginning of the game?* This question seeks to differentiate between mechanics that are given at the beginning of the game and mechanics that are acquired or introduced later.

3.1.2 *Utility.*

- *U1: Does the function of this mechanic vary with respect to its input?* This questions seeks to differentiate between mechanics whose outputs are static and mechanics with outputs that may vary with player input (e.g., by holding the jump button as opposed to tapping it, the player can jump higher).
- *U2: Could this mechanic be used to accomplish a variety of distinct goals?* This questions seeks to differentiate between mechanics that are highly contextual, and those that can be used to accomplish a variety of tasks (e.g., if the player can attack, they may be able to attack both objects in the game world and non-player characters).
- *U3: Could this mechanic be combined with another such that the combination of mechanics can achieve a distinct goal?* This question seeks to establish if the given mechanic has combinatorial reach–or if it can be combined with another mechanic to accomplish a goal that both mechanics, by themselves, could not.

3.1.3 *Impact.*

- *I1: Could this mechanic have a direct effect on the game world?* This question establishes whether or not the state of the game world, including player state, changes as a direct result of interaction with the mechanic.
- *I2: Could this mechanic have a direct effect on the game's narrative?* This question establishes whether or not the game's narrative, player-driven or well-defined, changes as a direct result of interaction with the mechanic.
- *I3: Could this mechanic have a direct effect on other mechanics, or on itself?* This question establishes whether or not gameplay changes as a direct result of interaction with the mechanic.

3.2 Constructing a Dataset of Game Mechanics

We built a dataset of game mechanics (n=47) from four different titles, each of a distinct genre. The games analyzed in this dataset were the following: *Super Mario Bros.*, *Dark Souls*, *Civilization 6*, and *Baldur's Gate 3*. We recorded the genre of each game, respectively: 2D Platformer, Action RPG, Turn-based Strategy, and Computer RPG. Mechanics were selected to be distinct from others, but there was a small degree of overlap. We also note here that the mechanics selected do not comprise an exhaustive list of mechanics from each game, with the exception of *Super Mario Bros.*, as it has relatively few mechanics.

In addition to these basic descriptors, we borrow one property from work done by Junior and Silva [5], wherein they decompose game mechanics into three categories: implied, core, and extra. Each game mechanic was ascribed this property according to the definitions laid out in their work. Following this, we evaluated each game mechanic according to our domain-specific questions. Each question reveals a binary property of the game mechanic within context of the game. As such, we relied on domain expert knowledge of these games to rate and validate the answers to these questions.

3.3 Statistical Analysis

We then performed a statistical analysis of our completed dataset. Each mechanic was scored along the three ontological domains (yes=1, no=0) and given a corresponding subscore for that domain. The minimum subscore for each domain is 0, while the maximum subscore for the availability domain is 4, and both utility and impact have maximum subscores of 3. An aggregate score–the sum of the domain subscores–was then given to each mechanic, ranging from 0-10. Additionally, we computed the average aggregate score by game, and, subsequently, by genre.

We then computed a set of pairwise correlations. As the properties of interest are nominal variables, we use the bias-corrected

Table 1: Summary Statistics of Domains

Domain	Mean	Median	Mode	Range
Availability	1.89	2	3	[0,4]
Utility	2.15	3	3	[0,3]
Impact	1.81	2	1	[0,3]
Aggregate	5.85	6	6	[2,10]

Table 2: Mean Domain Scores by Game

Game	A	U	I	Agg
Super Mario Bros.	2.75	1.75	1.25	5.75
Dark Souls	2.00	1.54	1.23	4.77
Civilization 6	1.00	2.58	2.25	5.83
Baldur's Gate 3	2.22	2.39	2.06	6.67

version of Cramer's v to measure the association between two properties and a χ^2 test to determine if the association is significant. We were also interested in examining the relationship between the mechanic type and the scores of its domains. For this measure, we use a simple regression fit of one variable against the other to determine the R^2 value, and record the F-statistic to determine significance. All statistical tests described in this work were performed with a significance level of $\alpha = 0.05$.

4 RESULTS

4.1 Summary Statistics

Table 1 denotes the mean, median, mode, and range of scores for each domain, with the aggregate score included. Of particular note with regards to these statistics is the fact that, across all domains the mean falls below the median score, suggesting that the distribution of scores in each case is negatively skewed. We also note that, disregarding the aggregate, utility has the highest mean score. This can be interpreted to mean that, generally, mechanics were rated as having more utility than either availability or impact.

To examine any differences between the games from which we selected mechanics, we computed the mean scores of each domain by game. Table 2 illustrates the result of these computations. We use the abbreviations A, U, I, and Agg in Table 2 to denote the domain being examined (availability, utility, impact, or aggregate). Before computing this set of statistics, we had expectations that the lowest mean aggregate score would belong to *Super Mario Bros.*, given that the game has relatively few mechanics and was designed many years before player agency became a point of study. This prognostication was, however, proven false with *Dark Souls* having the lowest score. We note here, however, that the difference in number of mechanics selected from each game may have had an impact on this result. Given that Computer RPGs are generally known for their breadth of mechanics and focus on player choice, we were unsurprised that *Baldur's Gate 3* showed the highest mean aggregate score.

We then looked at the distribution of mechanics according to type: implied, core, or extra. Implied mechanics made up around 64% of the dataset, with core mechanics making up the remaining 36% percent. No mechanic in the dataset was labeled as extra. We believe this skewed distribution of type favoring implied mechanics did have an impact on the analysis of this variable's relationships to the others.

Table 3: Highest/Lowest Aggregate Scores by Mechanic

Mechanic	Game	Genre	Agg
Explore map	BG3	Computer RPG	10
Interact with NPCs	BG3	Computer RPG	9
Use item	DS	Action RPG	8
Backstab (combat)	DS	Action RPG	2
Parry (combat)	DS	Action RPG	2
Upgrade equipment	DS	Action RPG	2

In pursuit of answering our third research question, we list the top three game mechanics with the highest aggregate score alongside the game from which they were selected, the game's genre, and their score in Table 3. We use the abbreviations BG3 and DS to denote the games *Baldur's Gate 3* and *Dark Souls*, respectively. According to our evaluation, exploration (BG3), interacting with NPCs (BG3), and using items (DS) were the mechanics with the highest aggregate score. These mechanics all reflect an inherent open-endedness of purpose, and within the context of each game, it was unsurprising that they ranked highly. For comparison, we also list the bottom three game mechanics in the same table. The backstab, parry, and upgrade equipment mechanics of DS were the mechanics with the lowest aggregate score. Given that the backstab and parry mechanics are highly contextual combat mechanics, they scored poorly in utility. Upgrading equipment, meanwhile, scored poorly in availability.

4.2 Pairwise Correlations

We plot our pairwise correlation assessment in Figure 1. Following previous standards, we use the abbreviations AScore, UScore, IScore, and AggScore to denote domain scores and aggregate score.

Of immediate note in the plot are the two negative correlations, shown by the blue squares, between the variable pairs AScore-UScore and AScore-IScore. Both relationships were determined to be insignificant at our chosen level of significance with p-values of .22 and .31, respectively. Despite these results, we believe these negative correlations can be explained to some degree through a design-focused lens. If a mechanic has such a large range of utility or impact on the game, designers of that game may wish that it should not be invoked so frequently or freely.

The relationship between Type and AggScore was shown to be significant ($p = .024$) with a weak association of .33. The coefficient for this variable when Type = Implied was 1.149, suggesting that the aggregate score of a mechanic increases by 1.149 when the mechanic is implied. We found this result to be unintuitive, as we expected core mechanics to allow for a higher degree of agency. We attribute this finding to the skewed distribution of mechanic type, as we noted in our summary statistics.

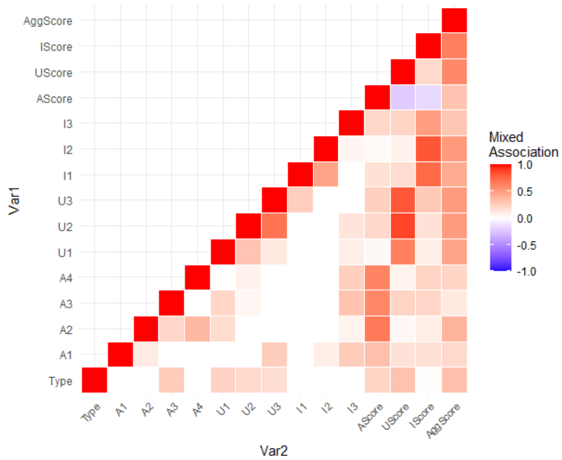

Figure 1: Pairwise correlation plot between individual properties, mechanic type, domain scores, and aggregate score. The level of association is shown by color scale, ranging from -1 (blue) to 1 (red).

Finally, we turn toward the relationships among the properties themselves. There is a distinct lack of detectable association between many properties at all, shown by the white squares in our plot. We interpret this finding as a positive thing. As far as a linear relationship is concerned, the properties we have selected seem to be moving relatively independent of each other. While we have not performed nonlinear tests to detect other kinds of relationships, we believe nonlinearity is unlikely given the form of our data. The pair of properties that had the highest association (.69) was U2-U3. This relationship was found to be statistically significant ($p < .001$). Given that both properties focus on determining goal-oriented utility of a mechanic, we found this result intuitive.

5 CONCLUSION

In this work, we have presented a preliminary study that builds toward an agency-centered ontology of game mechanics. We began with the following three research questions in mind: 1) How do we define agency-centered domains of mechanics, and what questions can we ask that best reveal properties of mechanics related to player agency? 2) How are these domains and properties interrelated, and are there domains or properties that contribute more to player agency than others? 3) What are extant examples of game mechanics that, according to our analysis, inspire the most player agency?

To answer our first research question, we defined a set of three agency-centered domains: availability, utility, and impact. We then devised a set of questions belonging to these domains meant to reveal binary properties of game mechanics. In order to answer our second and third research questions, we constructed a dataset of game mechanics across four games of different genres and evaluated those mechanics using our domain-specific questions. We scored those mechanics with regards to their domains and then analyzed this completed dataset. A summary analysis allowed us to offer specific examples of game mechanics that, according to our set of domains and scoring methodology, inspire the most agency. Finally, we explored the relationship between our domain-specific properties and other variables of interest, like mechanic type and the overall score given to the mechanic. We believe that game studies researchers, designers, and engineers alike can find benefit in a better understanding of game mechanics from an agency-centered perspective.

5.1 Future Work

This work is a preliminary step to building an agency-centered ontology of game mechanics. There are many directions in which this research can be taken, which we comment on here. The questions we devised belonging to the domains of availability, utility, and impact are not exhaustive, and we believe that both our set of domains and questions can be broadened with continued research into player agency theory and game design studies. Our dataset of game mechanics can be grown with regards to genre and game representation to provide a stronger sample and offer clearer statistical results. The dataset could also stand to benefit from outside validation by way of user study. This could, however, prove to be a difficult task, as it would require a pool of participants who are well-versed in all games and their mechanics belonging to the dataset. Finally, the results of this study could be used to engineer novel implementations of mechanical systems. In particular, we ask: What intelligent systems can be leveraged on mechanics to offer more agency, and spark joy in the players of our games? We intend to pursue these directions of study, and encourage our colleagues across disciplines to do the same.

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